

## STAGES OF A GRAPHIC CARD WORKING IN A SYSTEM

At its most basic, a graphic card creates an image that you see on your screen from binary data that it receives from the CPU. This data is converted to an image in parts. First, wireframes are created out of straight lines, and elements like texture, lighting and colour are added. The largest part of this work is done by the GPU, or graphic processing unit, a unit that compliments the CPU that is responsible for pixel and texture crunching among other things. There are other components on the graphic card other than the GPU, video RAM is one such component, as well as a video-decoding chip enabling playback. A graphic card also has a lot of power circuitry and power controllers, for like any PC component, GPUs need precise power delivery.

Graphic cards communicate with the CPU via a bus system on the motherboard. The current bus is PCI Express. As the need for more powerful cards arose, the need for faster, more efficient communication pathways between CPU and GPU also became more urgent. Therefore the AGP bus gave way to PCI Express, and currently we're at PCIe v2.0 with the next generation PCIe v3.0 boards already in circulation.

We've already established that the GPU receives data in the binary form from the CPU. This data could be about a simple bitmap image with colour references to each pixel, or a complex 3D game, in which case there will be changing pixel information, texture information, geometry information, shader information, as well as lighting information.

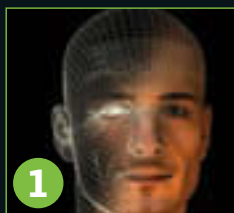
In the past, CPU sent all data to the GPU which was only required to draw an output on the display and nothing more. But nowadays, apart from getting graphical data, the GPU also helps in certain computing tasks by working in tandem in a heterogeneous computing model with the CPU. For instance, offloading certain tasks from CPU which can be performed better & faster using parallel processing in the GPU (such as CUDA cores) the tasks range from transcoding to molecular dynamics and computational finance and many more. This is also known as GPU Computing or such a setup is known as a GPGPU computing..

So let's look at what happens when you hit that game.exe file sitting on your desktop.

The GPU itself performs quite a few tasks as we've listed. These tasks are completed in what is referred to as a graphics pipeline. This pipeline is what converts all this 3D data into an image viewable on a 2D screen – your monitor. Both NVIDIA and AMD have slightly different pipelines, but the processes and stages are roughly the same, with a few unique steps in the form of tweaks, optimisations and such. Graphically or more accurately in terms of 3D rendering, these stages are shown alongside.

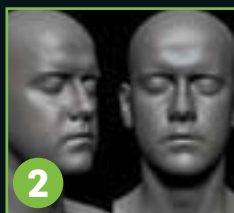
In case you are using an analogue display, the RAMDAC present on the graphic card converts this digital data to an analogue signal to be sent to the monitor via a D-Sub cable. In you are using a digital connectivity, such as an LCD monitor, there is no digital to analogue conversion required. 

## STAGES IN GPU WORKING



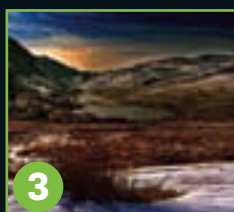
### TRANSFORMATION

This stage involves computing on vertex data, which are points in 3D space and converting these numbers into polygons in a 2D space. Based on the coordinates, these objects are placed relative to each other in what is a 2D space, although the objects remain 3D.



### LIGHTING

These polygons are lit correctly, as they will appear based on the positioning and number of light sources and reflective surfaces.



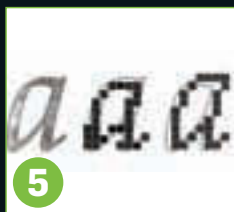
### VIEWPOINT

The viewpoint or orientation of the user is taken into account, closer objects will be larger and vice versa and all objects in the scene are manipulated on the basis of the orientation.



### CLIPPING

This involves pruning off the objects that will not be visible to the user. This saves the GPU precious computation power as calculations for objects not in the viewable scene need not be done.



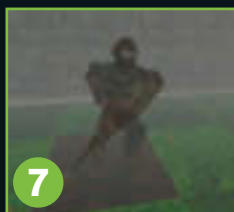
### RASTERIZATION

Objects are assigned to pixels, or in other words the scene is visualised as it would be on your monitor – a bunch of 2D pixels.



### TRIANGLE SETUP

More complex shapes are made from joining the vertex points, and creating multi-sided wireframe models as are seen in 3D scenes.



### CULLING

Non-viewable pixels are removed. This is actually the second stage of culling after clipping.



### TEXTURE AND SHADING

This involves performing a makeup job on each pixel. A texture is a 2D image that is added to the surface of a 3D object giving it a realistic look, while a pixel shader involves defining properties like colour, texture and depth effects to a pixel to give it a special appearance that mimics reality.